BAOMS
THE FACE OF SURGERY

# ORAL AND MAXILLOFACIAL SURGERY

## An Illustrated Guide for **Dental Students and Allied Dental Professionals**


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# PREFACE

Oral and Maxillofacial Surgery (OMFS) is the exciting specialty combining dentistry with medicine and surgery leading to a uniquely specialised and rewarding career which combines this knowledge with the ability to change and improve patients' faces, whether suffering pathology such as oral and pharyngeal cancers, improving dentofacial disproportion, treating children with congenital abnormalities such as cleft and craniofacial syndromes, TMJ and salivary gland pathology or performing facial aesthetic procedures.

Enhancing a smile can be very rewarding in dentistry. The face is our way of recognising who we are in the mirror and how we portray ourselves to others and any change has significant psychological impact and affects our confidence and success in life. The surgeon with an ability to change a face carries thus significant responsibility, but equally achieves greater professional satisfaction when a job is done well and a patient's quality of life is improved.

Most dental students have some exposure to OMFS via Oral Surgery and for those fortunate enough to spend some time with an OMFS team, many are inspired to go on to complete a shortened medical degree and pursue specialty training. OMFS surgeons work closely with allied surgical specialties such as neurosurgeons, ENT, Plastics and Ophthalmology colleagues but also the dental specialities in multidisciplinary teams with orthodontists, prosthodontists and periodontists and the scope of work includes the full remit of Oral Surgery.

The British Association of Oral and Maxillofacial Surgeons (BAOMS) has funded this book to widen awareness of what OMFS surgeons do and to give dental students and allied dental health professionals background knowledge on the scope of OMFS with practical examples of what we do. For the reader, it will give you an idea of emergency and urgent conditions and the likely options for provision of care for your patients when you refer them to us. I hope the book goes further and inspires you to touch base with your local OMFS unit or our dedicated colleagues at BAOMS via our website so we can support your career aspirations if you wish to learn more about OMFS.

**Miss Daljit K Dhariwal**
**62nd President BAOMS**

# ABOUT THE AUTHORS

**Nabeela Ahmed**
BDS, MFDS, MBChB, FRCS (OMFS), MMedSci (Med Ed)

Nabeela is a Consultant maxillofacial surgeon based at the Queens Medical Centre in Nottingham. She completed her registrar training across the London and Trent deaneries, and then undertook a 2 year post CCT fellowship in Australia and New Zealand with a trauma and skin cancer interest. Her current clinical practice is facial deformity, both acquired and post traumatic.

She has an active interest in medical education and has authored several textbook chapters in OMFS. She is also an elected fellow on BAOMS council.

**Rhea Chouhan**
BDS, MFDS, MBChB

Rhea is a dual qualified clinical fellow in Oral and Maxillofacial Surgery (OMFS), currently working in London. She has previously held elected positions as Chair of the Junior Trainees Group of The British Association of Oral and Maxillofacial Surgeons (BAOMS) and Junior Trainees & Members' Representative on BAOMS Council.

Having worked with BAOMS for a number of years, she is passionate about improving access and exposure to OMFS to both medical and dental undergraduates and in retaining current junior trainees by increasing learning and training opportunities. She has authored several papers in OMFS and collaborated with others to create a national mentorship scheme and hopes to continue supporting such projects in future during her training

**Robert Isaac**
BDS (Hons), MBBCH (Hons), FRCS (OMFS)

Robert has recently been appointed a Consultant in Oral and Maxillofacial surgery (OMFS) in Queen Alexandra Hospital, Portsmouth. He completed both his dental and medical degrees in Cardiff University before being appointed as 'run through' OMFS trainee in the Wessex Deanery.

He has written chapters in the 50 Landmark Papers in OMFS and Bailey and Love Essential Operations in OMFS and has become a reviewer for the British Journal of Oral and Maxillofacial Surgery. He has a particular aspiration to increase exposure to OMFS to both dental and medical undergraduates and holds an active interest in recruitment and retention within the specialty.

**Arpan S. Tahim**
MBBS, BSc (Hons), BDS, MRCS, MEd, PhD

Arpan is an honorary clinical lecturer at UCL medical school. He completed his PhD at the UCL Institute of Education. He has authored books, chapters and research articles focused on academic and clinical aspects of Medical Education and has set up numerous novel and successful educational interventions. His special interest lies in healthcare-related workplace learning and assessment.

**Karl F.B. Payne**
BMedSci, BMBS, BDS, FRCS, PhD

Karl is a Specialist Registrar in Oral and Maxillofacial surgery (OMFS) in the West Midlands Deanery, and NIHR Clinical Lecturer at the University of Birmingham. He completed his medical training in Nottingham and Dental training in London. As an academic surgeon, Karl's primary interests are using a blood test to detect DNA mutations from head and neck cancers. Karl has written several chapters and books in OMFS, raising awareness of the specialty at a national and international level.

**Alexander M.C. Goodson**
BSc (Hons), MBBS, BDS, FRCS (OMFS), DOHNS, PhD

Alex works as a Consultant maxillofacial surgeon at Queen Alexandra Hospital, Portsmouth and is an elected fellow on the BAOMS council. Having completed his specialty training in South Wales and a post-CCT Head and Neck fellowship in Birmingham, his subspecialty interests lie in head & neck oncology, microvascular reconstruction, salivary gland disease and facial nerve injury. He also has an active interest in facial aesthetic surgery and full-mouth dental implant rehabilitation.

He has a complementary clinical research interest in the continuing development of reverse planning and additively manufactured (customised 3D-printed) CMF implants for complex reconstructions with a focus on improving surgical efficiency and quality-of-life outcomes. He has taken an active role in raising the awareness of the specialty, having authored texts for primary care practitioners, as well as regularly teaching for the Midlands and East Deaneries and authoring texts for grass-roots trainees.

**Peter A. Brennan**
MB BS, BDS, MD, PhD, FRCS (Eng), FRCSI, Hon FRCS, FFST (Ed), FDSRCS

Professor Peter Brennan is a Consultant in Oral and Maxillofacial Surgery (OMFS) at Portsmouth, UK. He has a personal chair in recognition of his education and research profile.

Peter has over 830 publications with more than 80 on human factors (HF) and patient safety. He was awarded a PhD in HF in 2019 and as a trainee, completed an MD thesis on oral cancer biology. He is lead editor of several specialty surgical textbooks including the renowned Gray's Surgical Anatomy, the two volume Maxillofacial Surgery, Oxford Handbook of OMFS and the newly published Bailey and Love's Essential Operations in OMFS.

In 2022, Peter won the coveted Silver Scalpel Award for outstanding commitment to training across all 10 surgical specialties and was awarded an Honorary Fellowship of the Royal College of Surgeons for his achievements in wider surgery. His extensive HF work has helped changed practice, improve team working and patient safety around the world.

Peter is committed to training, supporting and empowering the future generation across wider healthcare.

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# CHAPTER 1: THE BASICS

## 1.1 INTRODUCTION

Oral and maxillofacial surgery (OMFS) originally developed during the middle of the twentieth century when dental surgeons began to apply their knowledge of the oral cavity and its surrounding tissues to treat military personnel who had sustained potentially devastating facial injuries during various conflicts, including the Second World War.

Since then, the remit of the speciality and the range of various pathologies within it have become increasingly diverse. As cases in the speciality became more complex, it was clear that in addition to a detailed knowledge of the oral cavity and facial skeleton, a firm grounding in the principles of both medicine and surgery was required to best manage these patients. So, in the late 1980s dual qualification in both dentistry and medicine became mandatory.

As dental students, you will be familiar with oral and maxillofacial surgery, as you will have exams in this speciality and, depending on where you train, may even do clinical blocks away from your base hospital to gain an insight into the work of OMFS surgeons. These placements may be the first time you are exposed to the speciality and become interested in finding out more about it, or even consider it as a potential career. Like many other surgical specialities, OMFS shares a tremendous amount in common with the other disciplines across both medicine and surgery. We will showcase the diversity of the speciality with this book by introducing you to OMFS and highlighting the overlap it shares with other specialities. We will take you through the different regions of the face and the different pathologies that patients present with, and introduce ways that OMFS can manage them. While it is by no means exhaustive in its scope and it cannot include all the pathology encountered in this complex anatomical region, we hope this book will inspire you to find out more about the problems that patients can develop in this region of the body.

The impact that disease and treatment in this area can have on patients and their families cannot be underestimated, as it is difficult to conceal the face from the outside world. We hope that the information and knowledge in this book will be useful for all students, whatever speciality or branch of dentistry you decide to pursue – whether that be in general dental practice or one of the many dental sub-specialities including oral-surgery/orthodontics/paediatrics/special-care dentistry, oral pathology, medicine or radiology. Or even, perhaps, OMFS itself.

Please do let us know if you think we should have included other topics too. The purpose of this book is to educate and support the knowledge of those looking after patients daily, and even to inspire some of you considering becoming OMFS surgeons to do so.

## 1.2 EXAMINING THE FACE AND MOUTH

As with all dental and medical presentations, the cornerstone of management of oral and maxillofacial pathology lies in obtaining a thorough history and examination. One of the challenges to examining the head and neck region is the need for a flexible approach. Your examination is often tailored to the situation, or the information you have picked up from your patient during the history. Similarly, how you examine a trauma patient might appear very different to how you examine a patient with a parotid swelling. However, the following framework should act as a foundation of general principles that you can adapt as you gain more experience.

Many clinical examinations begin by stressing the importance of noting any abnormalities affecting the head and neck. Building on that, we shall take you through a simple examination of the facial skeleton, the dental hard tissues, the oral soft tissues and the neck.

### Facial Examination

Simply observing the face at rest allows you to assess for facial asymmetry arising from, for example, facial swellings or muscle weakness. You can make a note of any lesions, bruising or lacerations. It is also useful to see how the jaw opens and closes – which reveals important information about the muscles of mastication and the temporomandibular joint (TMJ) itself. For example,

*Figure 1.1: Simply examining the patient at rest can help identify obvious deformity. This patient presented with an obvious depressed left frontal bone structure. Picture courtesy of Lieutenant Colonel Johno Breeze*

patients might show painful or restricted mouth opening, or even deviation of the jaw to one side as they open.

You can palpate the prominences of the bony skeleton – the orbital rims, the nasal bones, the cheeks (zygoma), the temporomandibular joints (TMJ) (where you can feel crepitus in an abnormal jaw joint) and the mandible itself. As you feel, you want to note if the area is tender, or if there is an underlying bony step that might suggest a facial bone fracture. Sometimes useful information is gained by feeling the bones inside the mouth. For example, midface fractures can be detected by simply grasping the maxilla on either side of the upper teeth and feeling if it is mobile (with your second hand stabilising the frontal bone). As part of the palpation process, remember to check the three divisions of the trigeminal nerve (sensation to the upper, middle and lower parts of the face, and motor supply to the masticatory muscles).

In the trauma situation, it is also useful to screen the eye – often by checking its gross appearance, gross vision, symmetry with the other eye, pupillary response and the range of eye movements. If you do this routinely, any abnormality will be easily picked up.

## Intra-oral Examination

Generally, the oral cavity can be examined without any specialist equipment, but it is imperative to have good lighting, and useful to have either a wooden tongue spatula or a dental mirror to hand.

An adult with a full dentition can normally have up to 32 teeth – occasionally more with supernumerary teeth. The tooth-bearing areas are divided into the four quadrants, each containing from the midline outwards: a central incisor, lateral incisor, a canine, two premolar teeth and three molar teeth. The last of these molar teeth are called 'wisdom teeth', but can be absent in some patients. Depending on the nature of the presenting complaint, important things to assess in a patient's teeth are:

a) Any significant dental decay (or caries) – often in the form of physical defects on the teeth or brown discolouration.
b) The presence of composite (usually white) or amalgam fillings and crowns.
c) Any mobility of the teeth, which can be done by simple palpation.
d) Any tenderness to percussion (often done by tapping the top of the tooth with an instrument; if this is painful, it often indicates inflammatory pathology at the tip of the tooth root).
e) Any discharging sinuses on the gums, which are a sign of chronic dental infection.

f) How a patient's teeth bite together (also called their '*dental occlusion*') as they dynamically open and close their jaw. As an example, patients with mandibular fractures often report a change in their bite, as the continuity of their dental arch is disrupted. Often you will see a gap between their teeth, or a step in their occlusion. This evaluation of the occlusion is also very relevant when considering whether patients have an underlying developmental skeletal discrepancy and may require orthodontics with orthognathic surgical input (we'll discuss this later, in Chapter 3).

g) Any absent teeth – and are they accounted for (especially in the trauma situation). For example, a tooth may have been avulsed; could it have been inhaled? Does the patient require a chest x-ray for evaluation?

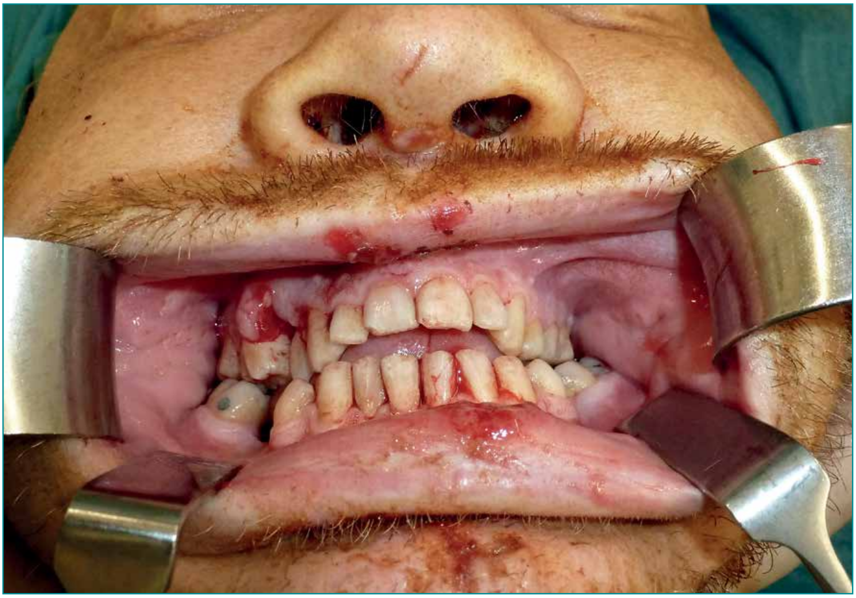

*Figure 1.2: This gentleman presented with an altered occlusion. Notice how his front teeth are wide open, despite biting on his posterior teeth. This is called an 'anterior open bite' or 'AOB'. This is commonly seen in patients with Le Fort fractures. Picture courtesy of Mr. Madan Ethunandan*

It is important to systematically assess the mucosal surfaces. These areas include:

- The buccal mucosa (inner lining of cheek), including the opening of the parotid duct opposite the upper second molar

*Figure 1.3: The parotid duct opens in the buccal mucosa opposite the upper second molar. Here it can be seen to discharging frank pus. This is usually indicative of parotitis. Picture courtesy of Mr. S Walsh*

- The labial mucosa (inner surface of lips) and frenulum, which can be injured
- The lips themselves, including the presence of any numbness/altered sensation
- The gingival surfaces (gums) – including looking for gum boils (sinuses), as mentioned above

*Figure 1.4: You can see that the gingivae of the lower front teeth looks acutely inflamed, with evidence of desquamation or 'sloughing'. This patient was being treated for pemphigus, a blistering condition. Picture courtesy of Mr. Stephen Walsh*

- The hard palate
- The soft palate
- The tongue (dorsal surface, lateral border and ventral aspect)
- The floor of the mouth
- The oropharynx
- The retromolar region (area of mucosa just behind the last standing molar tooth).

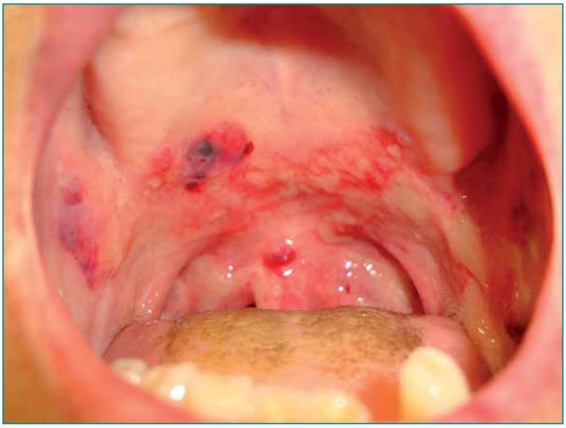

*Figure 1.5: Here at the junction between the hard and the soft palate, the patient presented with acute blistering. Notice that on the right soft palate some of the blisters look like they are blood filled. This was mucous membrane pemphigoid. Picture courtesy of Mr. Stephen Walsh*

*Figure 1.6: This patient presented with a mass in the right floor of the mouth. This was later found to be a salivary stone, which is just visible beneath the mucosal surface. The patient also presented with mandibular tori, bony exostoses that will be covered later in this book. Picture courtesy of Mr. Stephen Walsh*

As with any lump or bump, any lesion (for example, a swelling or ulcer) found in the mouth should be assessed to establish its site, size, shape, colour and border, and for the presence of any associated (cervical) lymphadenopathy.

## Neck Examination

The principles of a good neck examination will be familiar, and the standard inspection and palpation sequence should be followed. Observation might highlight any areas of concern, and examination of lumps and lesions can be described in terms of their site, size, shape, colour, consistency, nature of their borders and any trans-illumination (though this is rarely used in the head and neck). They are also assessed in relation to swallowing and tongue protrusion.

If a neck examination reveals lymphadenopathy, it is imperative to examine the skin, oral cavity, nose and face for potential sources of malignancy, and where appropriate, the other lymph-node sites (axilla, groin) and abdomen for signs of other lymphadenopathy or hepato-splenomegaly. We'll review this again in Chapter 5 where we look at neck lumps.

*Figure 1.7: A patient presented with a change in a mole. Excisional biopsy identified this as malignant melanoma. Whilst less common than other forms of skin cancer, melanoma can present with lymphadenopathy in the head and neck. Picture courtesy of Mr. Stephen Walsh*

Of course, depending on your findings, you might need to carry out more detailed examination of particular areas – for example, the eye, ear or cranial nerves. However, the examination framework above will hopefully be useful to help you better understand the information in the next chapters.

*Figure 1.8: This patient was referred in with a right neck lump. A mass was identified in the right tail of the parotid. Most parotid masses are benign. Picture courtesy of Mr. Stephen Walsh*

## 1.3 SOME IMPORTANT INVESTIGATIONS

Depending on your history and examination findings, further information is often gained through a variety of further investigations, ranging from simple blood tests to complex forms of imaging. Not all of these will be immediately available to you in a dental setting, and may require referral to the hospital directly or via the patient's GP.

Head and neck imaging is complicated to interpret. As dental students, you will tend to be unfamiliar with soft-tissue imaging but will be well versed in the interpretation of the OPG and intra-oral periapical films. Increasingly, the use of cone beam CT scans (CBCT) is an area of learning for dental students and dental practitioners alike.

## Plain Film Imaging

Below are examples of the three most common forms of plain radiograph image that help us better investigate and visualise the facial skeleton.

*Figure 1.9: This is the most common radiograph used in OMFS. It is called an 'orthopantomogram'. It is a panoramic radiograph of the maxilla and mandible, showing all of the teeth, skeletal structures, maxillary sinuses and temporomandibular joints. It is useful for assessment of dental caries, abscesses, potential trauma or, as in this example, identifying submandibular stones. Picture courtesy of Mr. Stephen Walsh*

*Figure 1.10: This is a 'PA mandible'. This, when combined with an OPG, can help identify fractures of the mandible. Here a fracture of the left angle of the mandible can be seen.*